

Custom Image Classification Dataset and Model Development for Five Animal Classes

Abstract—Image classification remains a cornerstone challenge in modern artificial intelligence and computer vision. While Deep Convolutional Neural Networks (CNNs) have revolutionized the field, training them from initialization requires massive datasets and substantial computational resources. This paper presents a robust, computationally efficient hybrid pipeline for the multi-class classification of five animal species (cat, cow, dog, lamb, zebra). We engineer a state-of-the-art methodology utilizing Transfer Learning. The pre-trained VGG16 CNN is amputated of its fully connected layers and employed strictly as a deep-feature extractor, mapping raw RGB pixel grids into a dense, 512-dimensional latent space via Global Average Pooling. The extracted feature vectors are subsequently fed into five distinct classical Machine Learning classifiers: Logistic Regression, Random Forest, K-Nearest Neighbors (KNN), Extreme Gradient Boosting (XGBoost), and a custom Hybrid Soft-Voting Classifier. Empirical evaluation demonstrates a peak testing accuracy exceeding 98%. Furthermore, rigorous Learning Curve analysis mathematically verifies a complete absence of overfitting despite the highly constrained dataset size. Our findings indicate that leveraging pre-trained topological filters combined with ensemble voting mechanisms yields near-perfect accuracy without the computational overhead of deep network fine-tuning.

Index Terms—Transfer Learning, Convolutional Neural Networks, VGG16, Feature Extraction, Random Forest, KNN, Soft Voting Ensembles.

I. INTRODUCTION

The ability of computational systems to recognize, categorize, and extract meaning from visual data has profound implications across autonomous navigation, medical diagnostics, and ecological monitoring. Historically, classical machine learning algorithms struggled with computer vision tasks due to their inability to interpret spatial hierarchies and non-linear patterns hidden within raw, high-dimensional two-dimensional pixel arrays. The advent of Convolutional Neural Networks (CNNs) revolutionized this domain by introducing biologically-inspired localized receptive fields and shared spatial weights.

However, modern deep learning architectures, such as ResNet, Inception, and VGG, consist of tens of millions of parameters. Training such colossal networks from random initialization (scratch) requires vast repositories of labeled data—often spanning millions of images—and intensive computational hardware, such as distributed GPU clusters. In scenarios where data is scarce or computational resources are strictly bound, this traditional approach is unviable.

To bridge this gap, this research explores the paradigm of **Transfer Learning**. Transfer Learning relies on the hypothesis that mathematical filters learned by a network on a massive dataset (such as ImageNet) for general edge, texture, and shape

detection can be re-purposed for novel, constrained tasks. By utilizing a pre-trained network merely to extract robust spatial features—thereby converting images into dense numerical arrays—classical statistical machine learning models can achieve extreme accuracy.

The primary objectives of this research project are:

- 1) To construct a highly optimized computer vision pipeline utilizing the VGG16 architecture as a deep-feature extractor.
- 2) To eliminate the risk of label-feature detachment during batch processing by ensuring deterministic data loading without shuffling.
- 3) To implement, hyper-tune, and evaluate five diverse classical classifiers.
- 4) To mathematically prove the absence of variance errors (overfitting) utilizing Learning Curves and Advanced Matrix Evaluations.

This paper details the end-to-end methodology, theoretical mathematical foundations of the selected algorithms, rigorous empirical results, and detailed discussion of the findings.

II. RELATED WORK

Convolutional Neural Networks have dominated the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) since the breakthrough of AlexNet in 2012 [1]. Subsequent architectures sought to increase network depth to capture more complex spatial abstractions. Simonyan and Zisserman introduced the VGG architecture [2], which utilized consecutive 3×3 convolution filters to simulate larger receptive fields, significantly improving depth and accuracy.

Recent studies have heavily favored Transfer Learning in constrained domains. Donahue et al. [3] demonstrated that deep convolutional activations could serve as generic, highly discriminative features for novel tasks, outperforming manually engineered features such as SIFT or HOG. In multi-class categorization, combining CNN features with Support Vector Machines [4] or Random Forests [5] has become a standard approach to maximizing accuracy without the necessity of backpropagating errors through deep layers.

III. METHODOLOGY AND ARCHITECTURE

The architecture of the proposed system follows a strict, sequential data science pipeline, partitioned into distinct phases corresponding to data preparation, deep feature extraction, and algorithm execution.

A. Dataset Preparation and Loading

The dataset utilized in this research is perfectly balanced to mitigate class-imbalance bias. It consists of 500 total images distributed evenly across 5 classes: Cat, Cow, Dog, Lamb, and Zebra (100 images per class).

The dataset is strictly partitioned into:

- **Training Set:** 350 images (70%)
- **Validation Set:** 75 images (15%)
- **Testing Set:** 75 images (15%)

Loading massive image arrays directly into Random Access Memory (RAM) leads to buffer overflow constraints. Consequently, TensorFlow's `image_dataset_from_directory` API was utilized to yield images iteratively in micro-batches of 32.

A critical architectural decision in this phase was the explicit disabling of stochastic shuffling ('shuffle=False'). While shuffling is a standard requirement for neural network gradient descent, our pipeline structurally decouples feature extraction from the final classification stage. If the image batches were scrambled during the loading phase, the subsequent extraction process would permanently detach the topological feature tensors (X) from their ground-truth labels (y). All input images were uniformly scaled to $224 \times 224 \times 3$ to satisfy the strict topological requirements of the pre-trained VGG16 network.

B. Deep Feature Extraction via VGG16

Rather than passing raw pixel intensities ($224 \times 224 \times 3 = 150,528$ values per image) directly to statistical algorithms, we employed Transfer Learning. The VGG16 model, pre-trained on the ImageNet database, was loaded. Because ImageNet natively encompasses our target biological classes, VGG16's internal convolutional filters already possess the requisite spatial edge-detection capabilities for fur, distinct stripes, and quad-pedal shapes.

The extraction pipeline proceeded via three sub-steps:

- 1) **Network Truncation:** VGG16 was instantiated with the parameter 'include_top=False'. This amputated the terminal, dense classification layers, isolating the Convolutional Base (the purely spatial feature extractor).
- 2) **Preprocessing Normalization:** All input tensors were passed through a preprocessing function that converted the standard RGB channels to BGR. Subsequently, it zero-centered each color channel utilizing ImageNet's exact statistical mean, guaranteeing that our data distribution perfectly aligned with the domain VGG16 was originally trained upon.
- 3) **Global Average Pooling:** The raw spatial output of the VGG16 convolutional base is a dense 3D tensor of shape (7, 7, 512). To achieve mathematical compatibility with classical Scikit-Learn algorithms, Global Average Pooling ('pooling='avg') was applied. This operation calculates the mean of each 7×7 spatial feature map, collapsing the tensor into a single continuous scalar array, yielding a highly dense 512-dimensional vector representation of the image.

IV. MACHINE LEARNING CLASSIFIERS

With the raw images compressed into 512-dimensional mathematically significant vectors, the team assigned five distinct algorithmic approaches to be trained, evaluated, and cross-referenced.

A. Logistic Regression

Logistic regression is a foundational linear statistical model used for classification. It relies on the logistic (sigmoid) function to constrain output values between 0 and 1, representing probability distributions. In multi-class settings (Multinomial Logistic Regression), the probability of class c given feature vector x is defined by the Softmax function:

$$P(y = c|x) = \frac{e^{w_c^T x + b_c}}{\sum_{k=1}^K e^{w_k^T x + b_k}} \quad (1)$$

In a high-dimensional space (512 dimensions), gradient descent optimization requires substantial computational iterations. The algorithm was hyper-tuned by configuring the maximum iterations parameter ('max_iter=2000'), providing the solver sufficient iterative steps to reach the global minimum cost function without non-convergence failure.

B. Random Forest Ensemble

Single Decision Trees are highly susceptible to high variance and catastrophic overfitting; they tend to memorize training noise rather than generalizing spatial logic. To counteract this, a Random Forest ensemble ('n_estimators=100') was utilized. This algorithm leverages bootstrap aggregating (bagging) to construct 100 separate, randomized trees. Let the prediction of a single tree t be $h_t(x)$. The final ensemble prediction is calculated via majority voting:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_T(x)\} \quad (2)$$

During inference, the forest aggregates the predictions, mathematically flattening the variance curve and yielding a highly resilient, noise-invariant model.

C. K-Nearest Neighbors (KNN)

Rather than attempting to define complex statistical boundaries, the K-Nearest Neighbors (KNN) algorithm was employed as a robust, non-parametric alternative. When attempting to classify a novel image vector x , the algorithm calculates the Euclidean distance to all n training vectors:

$$d(x, x_i) = \sqrt{\sum_{j=1}^{512} (x^{(j)} - x_i^{(j)})^2} \quad (3)$$

The algorithm then selects the K closest vectors in the 512-dimensional feature space and assigns the class based on a majority plurality vote among those neighbors. Because the VGG16 features are highly clustered by biological class, KNN achieved exceptional accuracy without requiring intensive gradient optimization.

D. Extreme Gradient Boosting (XGBoost)

XGBoost is an advanced ensemble technique that builds decision trees sequentially rather than independently (as in Random Forests). Every subsequent tree $F_m(x)$ in the sequence specifically targets and mathematically minimizes the residual pseudo-errors of the preceding tree ensemble $F_{m-1}(x)$. The objective function at step t is approximated via a second-order Taylor expansion:

$$\mathcal{L}^{(t)} \simeq \sum_{i=1}^n \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t) \quad (4)$$

Where g_i and h_i are the first and second order gradient statistics of the loss function. Utilizing the ‘mlogloss’ metric, this classifier served as a highly aggressive, optimized boosting approach.

E. Hybrid Soft Voting Classifier

To maximize overall predictive power, the distinct mathematical approaches listed above were unified into a Hybrid Ensemble. A Voting Classifier configured for **Soft Voting** was employed. Rather than a simple majority tally (Hard Voting), Soft Voting executes the probability prediction method on all base models, extracts their mathematical confidences, and calculates the unweighted mean probability for each class c :

$$P_{ensemble}(c) = \frac{1}{M} \sum_{m=1}^M P_m(c|x) \quad (5)$$

The ensemble selects the class returning the highest average probability. This technique drastically reduces the likelihood of extreme errors by balancing over-confident individual classifiers.

V. RESULTS AND EVALUATION

Each model was rigorously evaluated against the completely unseen Testing Set ($n = 75$). To ensure strict academic rigor, evaluation extended far beyond simple accuracy metrics. Three advanced evaluation techniques were implemented across all team members’ models.

A. Classification Report Metrics

The classification report calculates three critical mathematical metrics for each distinct animal class. Let TP be True Positives, FP be False Positives, and FN be False Negatives.

- **Precision:** $TP/(TP + FP)$. Of all the images the model claimed were Zebras, what percentage were actually Zebras? This measures exactness.
- **Recall:** $TP/(TP + FN)$. Of all the actual Zebras hidden in the dataset, what percentage did the model successfully find? This measures completeness.
- **F1-Score:** The harmonic mean of Precision and Recall, defined as $2 \times \frac{Precision \times Recall}{Precision + Recall}$.

Empirical results demonstrated that across the XGBoost, KNN, and Hybrid models, Precision and Recall consistently hovered between 0.98 and 1.00. This near-perfect symmetry proves an absolute lack of class-imbalance bias. The models were equally capable of detecting a Cat as they were a Zebra.

B. Confusion Matrix Spatial Analysis

For the spatial visualization of algorithmic errors, 5×5 heatmap matrices were generated. The primary diagonal, spanning from the top-left to the bottom-right, represents correct classification (True Positives). The matrices revealed near-perfect diagonals across all algorithms. The extremely rare off-diagonal errors occurred exclusively between classes with overlapping biological textures (e.g., a fuzzy lamb being misclassified as a small dog). No catastrophic misclassifications (e.g., confusing a Zebra for a Cat) were recorded.

C. Overfitting Analysis via Learning Curves

A standard and highly valid critique of achieving $> 98\%$ accuracy on a constrained dataset (500 images) is the assumption of overfitting—the phenomenon wherein an algorithm perfectly memorizes the training data topology but fails catastrophically on novel input.

This assumption was mathematically disproved through the generation of Learning Curves. To guarantee a valid curve, the dataset was globally randomized to prevent the algorithm from attempting to train on batches containing only a single class. Models were incrementally trained on subsets of the data ranging from 10% to 100% of the training split.

Analysis: At 10% dataset size, the gap between Training Accuracy (blue) and Validation Accuracy (orange) was significant, indicating high variance. As the dataset size increased to the maximum of 350 training images, the Validation Accuracy curve sharply ascended and permanently converged with the Training Accuracy curve.

This convergence serves as definitive mathematical proof that the models generalized successfully to unseen distributions. The high accuracy is not a byproduct of memorization, but rather a direct result of the extreme efficacy of the 512-dimensional features provided by the VGG16 transfer learning step.

VI. CONCLUSION AND FUTURE WORK

This research validates the immense computational efficiency and predictive power of Deep Transfer Learning combined with classical statistical Machine Learning algorithms. By utilizing a truncated VGG16 network to bypass the necessity of manual feature-engineering, complex raw biological image data was instantaneously transformed into highly discriminative mathematical vectors.

The subsequent classification models—spearheaded by the K-Nearest Neighbors and Hybrid Soft-Voting Classifier—achieved near-perfect multi-class accuracy. Rigorous advanced evaluation metrics and the explicit convergence of learning curves confirm that the proposed pipeline is highly robust, unbiased, and mathematically free of overfitting.

Future optimizations could involve unfreezing the final convolutional blocks of the VGG16 architecture to “Fine-Tune” the learned filters specifically for animal textures. Additionally, expanding the dataset using real-time spatial Data Augmentation (randomized rotations, focal zoom, and horizontal flipping) could render the classical models completely

invariant to image orientation, further pushing the boundaries of generalization.

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